Download the Selenium Server from here.

Selenium Grid Tutorial: Step by Step Guide with Example
Step 2

You can place the Selenium Server .jar file anywhere in your HardDrive. But for the purpose of this tutorial, place it on the C drive of both Machine A and Machine B. After doing this, you are now done installing Selenium Grid. The following steps will launch the hub and the node.

Step 3

We are now going to launch a hub. Go to Machine A. Using the command prompt, navigate to the root of Machine A's – C drive, because that is the directory where we placed the Selenium Server.
On the command prompt, type java -jar selenium-server-standalone-2.30.0.jar -role hub
The hub should successfully be launched. Your command prompt should look similar to the image below
Selenium Grid Tutorial: Step by Step Guide with Example
Step 4

Another way to verify whether the hub is running is by using a browser. Selenium Grid, by default, uses Machine A's port 4444 for its web interface. Simply open up a browser and go to http://localhost:4444/grid/console

Selenium Grid Tutorial: Step by Step Guide with Example

Also, you can check if Machine B can access the hub's web interface by launching a browser there and going to where "iporhostnameofmachineA" should be the IP address or the hostname of the machine where the hub is running. Since Machine A's IP address is 192.168.1.3, then on the browser on Machine B you should type http://192.168.1.3:4444/grid/console

Step 5

Now that the hub is already set up, we are going to launch a node. Go to Machine B and launch a command prompt there.
Navigate to the root of Drive C and type the code below. We used the IP address 192.168.1.3 because that is where the hub is running. We also used port 5566 though you may choose any free port number you desire.
NOTE: You now have to give path to the Gecko driver if using Firefox. Here is updated code that needs to be used
java -Dwebdriver.gecko.driver="C:\geckodriver.exe" -jar selenium-server-standalone-3.4.0.jar -role webdriver -hub http://192.168.1.3:4444/grid/register -port 5566
Selenium Grid Tutorial: Step by Step Guide with Example

When you press Enter, your command prompt should be similar to the image below.
Selenium Grid Tutorial: Step by Step Guide with Example
Step 6

Go to the Selenium Grid web interface and refresh the page. You should see something like this.

At this point, you have already configured a simple grid. You are now ready to run a test remotely on Machine B.

Designing Test Scripts That Can Run on the Grid

To design test scripts that will run on the grid, we need to use DesiredCapabilites and the RemoteWebDriver objects.

DesiredCapabilites is used to set the type of browser and OS that we will automate
RemoteWebDriver is used to set which node (or machine) that our test will run against.
To use the DesiredCapabilites object, you must first import this package

To use the RemoteWebDriver object, you must import these packages.

Using the DesiredCapabilites Object
Go to the Grid's web interface and hover on an image of the browser that you want to automate. Take note of the platform, and the browserName showed by the tooltip.

In this case, the platform is "XP" and the browserName is "Firefox."

We will use the platform and the browserName in our WebDriver as shown below (of course you need to import the necessary packages first).

Using the RemoteWebDriver Object
Import the necessary packages for RemoteWebDriver and then pass the DesiredCapabilities object that we created above as a parameter for the RemoteWebDriver object.

Running a Sample Test Case on the Grid
Below is a simple WebDriver Testng code that you can create in Eclipse on Machine A. Once you run it, automation will be performed on Machine B.

The test should pass.

 Selenium Grid Tutorial: Step by Step Guide with Example
Selenium grid configuration using JSON File:

JSON stands for Javascript Object Notation. It is a standard format used for interchange of data between browser and the web server. Selenium has an in built JSON config file that can be used to set up selenium grid.

Below are the steps to configure selenium grid using JSON config file.

Step 1) Download the code for JSON config file using the below path

https://github.com/SeleniumHQ/selenium/blob/selenium-3.141.59/java/server/src/org/openqa/grid/common/defaults/DefaultHub.json

Here is the code

```
{
        "port": 4444,
        "newSessionWaitTimeout": -1,
        "servlets" : [],
        "withoutServlets": [],
        "custom": {},
        "capabilityMatcher": "org.openqa.grid.internal.utils.DefaultCapabilityMatcher",
        "registryClass": "org.openqa.grid.internal.DefaultGridRegistry",
        "throwOnCapabilityNotPresent": true,
        "cleanUpCycle": 5000,
        "role": "hub",
        "debug": false,
        "browserTimeout": 0,
        "timeout": 1800
}
```

Step 2) Copy and paste the code on a text editor such as Notepad with the extension '.json'

Step 3)Launch the hub using the below command using command prompt

java -jar selenium-server-standalone-2.53.1.jar -role hub -hubConfig hubconfig.json
NOTE: Selenium stand alone jar file and the json file must be present on the same directory

Step 4) The below screen will appear which indicates the hub is set up successfully

Step 5) Open web browser and try connecting to the below URL
http://192.168.43.223:4444/grid/console
NOTE: The URL may vary from machine to machine. URL followed by 'Nodes should register to' on the above screen must be used.

Step 6) Grid Console screen will appear which indicates the hub is up and running

Configure NODES using JSON:

Step 1) Download the sample node configuration file provided by Selenium using the below URL

https://github.com/SeleniumHQ/selenium/blob/selenium-2.53.0/java/server/src/org/openqa/grid/common/defaults/DefaultNode.json

NOTE: IF YOU ARE USING SELENIUM 3.0 OR ABOVE, THE BELOW JSON FILE MUST BE USED

https://github.com/SeleniumHQ/selenium/blob/selenium-3.141.59/java/server/src/org/openqa/grid/common/defaults/DefaultNodeWebDriver.json

Step 2) Copy and paste the code into a text editor such as Notepad and save the file with the extension '.json'

Step 3)

Run the below command on command prompt

Java-Dwebdriver.chrome.driver="chromedriver.exe" -Dwebdriver.ie.driver="IEDriverServer.exe" -Dwebdriver.gecko.driver="geckodriver.exe" -jar selenium-server-standalone-2.53.1.jar -role node -nodeConfig node1.json
NOTE: Before running the command, please make sure the drivers for each of the browsers have been downloaded onto local machine directory

Step 4) The below screen will appear when enter key is pressed

Step 5) Navigate to the grid console to verify if the nodes have been successfully configured

The browsers are displayed on the grid console. Hence we can be sure that nodes are configured successfully.

Sample Test Cases On Selenium Grid:

```
package com.objectrepository.demo;
import org.openqa.selenium.*;
import org.openqa.selenium.remote.DesiredCapabilities;
import java.net.MalformedURLException;
import java.net.URL;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.*;

public class SeleniumGrid {

    WebDriver driver;
    String baseURL, nodeURL;

    @BeforeTest
    public void setUp() throws MalformedURLException {
       baseURL = "http://demo.guru99.com/test/guru99home/";
```

```
        nodeURL = "http://192.168.43.223:5566/wd/hub";
        DesiredCapabilities capability = DesiredCapabilities.chrome();

        capability.setBrowserName("chrome");
        capability.setPlatform(Platform.WIN10);
        driver = new RemoteWebDriver(new URL(nodeURL), capability);
    }

    @AfterTest
    public void afterTest() {
        driver.quit();
    }
    @Test
    public void sampleTest() {
        driver.get(baseURL);


        if (driver.getPageSource().contains("MOBILE TESTING")) {
            Assert.assertTrue(true, "Mobile Testing Link Found");
        } else {
            Assert.assertTrue(false, "Failed: Link not found");
        }

    }

}
```

The above code launches chrome browser and navigates to the URL specified in the 'baseURL' variable. It verified a link name 'Mobile Testing' is displayed on the page

URL on the 'nodeURL' variable can be modified to reflect the IP Address of the remote machine.

Test result can be verified on the default TestNG report generated

elenium installation is a 3 step process:

Install Java SDK
Install Eclipse
Install Selenium Webdriver Files

In this tutorial, we will learn how to install Selenium Webdriver . Below is the detailed process

NOTE: The versions of Java, Eclipse, Selenium will keep updating with time. But the installation steps will remain the same. Please select the latest version and continue the installation steps below-

Step 1 – Install Java on your computer
Download and install the Java Software Development Kit (JDK) here.

How to Download & Install Selenium
WebDriver Next –

This JDK version comes bundled with Java Runtime Environment (JRE), so you do not need to download and install the JRE separately.

Once installation is complete, open command prompt and type “java”. If you see the following screen you are good to move to the next step

How to Download & Install Selenium
WebDriver Step 2 – Install Eclipse IDE
Download latest version of “Eclipse IDE for Java Developers” here. Be sure to choose correctly between Windows 32 Bit and 64 Bit versions.

You should be able to download an exe file named “eclipse-inst-win64” for Setup.

Double-click on file to Install the Eclipse. A new window will open. Click Eclipse IDE for Java Developers.

After that, a new window will open which click button marked 1 and change path to “C:\eclipse”.
Post that Click on Install button marked 2

After successful completion of the installation procedure, a window will appear. On that window click on Launch

This will start eclipse neon IDE for you.

Step 3 – Download the Selenium Java Client Driver
You can download Selenium Webdriver for Java Client Driver here. You will find client drivers for other languages there, but only choose the one for Java.

This download comes as a ZIP file named “selenium-3.14.0.zip”. For simplicity of Selenium installation on Windows 10, extract the contents of this ZIP file on your C drive so that you would have the directory “C:\selenium-3.14.0\”. This directory contains all the JAR files that we would later import on Eclipse for Selenium setup.

Step 4 – Configure Eclipse IDE with WebDriver
Launch the “eclipse.exe” file inside the “eclipse” folder that we extracted in step 2. If you followed step 2 correctly, the executable should be located on C:\eclipse\eclipse.exe.
When asked to select for a workspace, just accept the default location.

3. Create a new project through File > New > Java Project. Name the project as “newproject”.

A new pop-up window will open enter details as follow
Project Name
Location to save project
Select an execution JRE
Select layout project option
Click on Finish button

4. In this step,

Right-click on the newly created project and

Select New > Package, and name that package as “newpackage”.

A pop-up window will open to name the package,

Enter the name of the package

Click on Finish button

5. Create a new Java class under newpackage by right-clicking on it and then selecting- New > Class, and then name it as “MyClass”. Your Eclipse IDE should look like the image below.

When you click on Class, a pop-up window will open, enter details as
Name of the class
Click on Finish button

This is how it looks like after creating class.

Now selenium WebDriver’s into Java Build Path

In this step,

Right-click on “newproject” and select Properties.
On the Properties dialog, click on “Java Build Path”.
Click on the Libraries tab, and then
Click on “Add External JARs..”

When you click on “Add External JARs..” It will open a pop-up window. Select the JAR files you want to add.

After selecting jar files, click on OK button.

Select all files inside the lib folder.

Select files outside lib folder

How to Download & Install Selenium
WebDriver Once done, click “Apply and Close”
button

How to Download & Install Selenium WebDriver

6. Add all the JAR files inside and outside the “libs” folder. Your Properties dialog should now look similar to the image below.

7. Finally, click OK and we are done importing Selenium libraries into our project.

Different Drivers

HTMLUnit and Firefox are two browsers that WebDriver can directly automate – meaning that no other separate component is needed to install or run while the test is being executed. For other browsers, a separate program is needed. That program is called as the Driver Server.

A driver server is different for each browser. For example, Internet Explorer has its own driver server which you cannot use on other browsers. Below is the list of driver servers and the corresponding browsers that use them.

Conclusion: Hence we learned how to install selenium grid and selenium web driver.